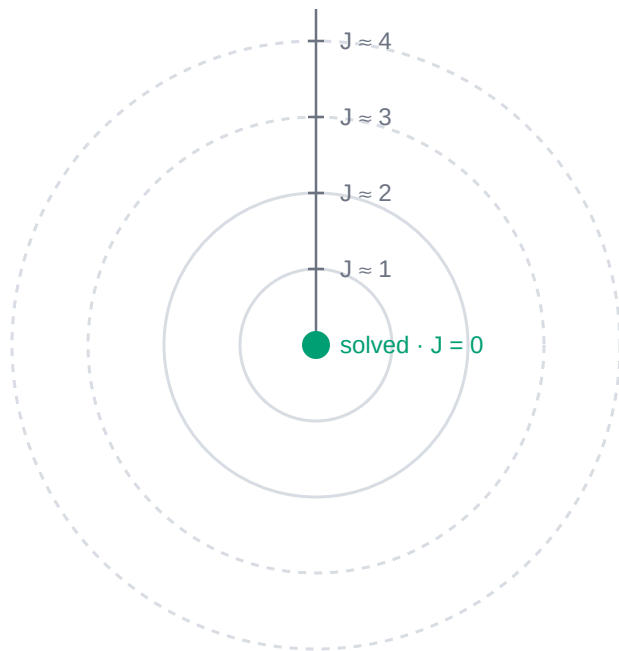


value spreads outward from solved



one shell per target-network refresh

The update

$J(s) = 0$ if s solved

$J(s) = \min_a 1 + J(s'_a)$ otherwise

Why the data is free

scramble k moves from solved $\rightarrow$ a labelled state.
no dataset to collect, no annotation, unlimited.

Why a curriculum

only states near solved have a grounded target.
sampling $k \sim U(1, 40)$ over 63 moves put 2.5% of the batch there; the run collapsed to a constant
 $J = 15.33$ at every depth. Growing k fixed it.

What it cannot give you

a learned heuristic is not admissible, so the result is near-optimal: never proven minimal.